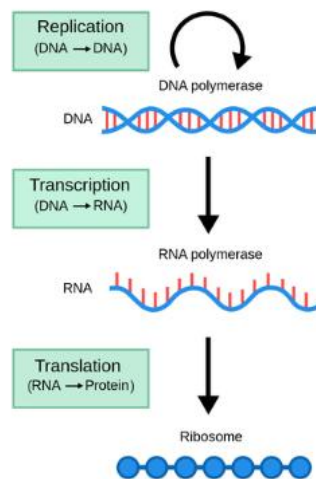


BIOINFORMATICS

WEEK 1 – DAY 4

SEQUENCE ALIGNMENT - It is a way of arranging the primary sequences of DNA, RNA or Proteins to identify regions of similarity that may be a consequences of FUNCTIONAL, STRUCTURAL and EVOLUTIONARY relationship between the sequences.



- Sequence alignment is the procedure of comparing two (pair-wise alignment) or more (multiple sequences) by searching for a series of individual characters/patterns that are in the same order in the sequences.
- This can also be defined as “ the alignment is made between a known sequence & unknown sequence or between 2 unknown sequence”.

Here, the known sequence is called the “**REFERENCE SEQUENCE**” and the unknown sequence is called the “**QUERY SEQUENCE**”.

Sequence alignment is the process of placing sequences side by side to identify:

- Similarities
- Differences
- Mutations
- Conserved regions
- Evolutionary relationships

TYPES IN SEQUENCE ALIGNMENT

- **PAIRWISE ALIGNMENT** : It is a method used to find the best-matching piece-wise(Global & Local) alignments of 2 query sequence at a time.
- **MULTIPLE SEQUENCE ALIGNMENT** : It is an extension of pairwise alignment to incorporate 3/ more sequences of similar length at a time.

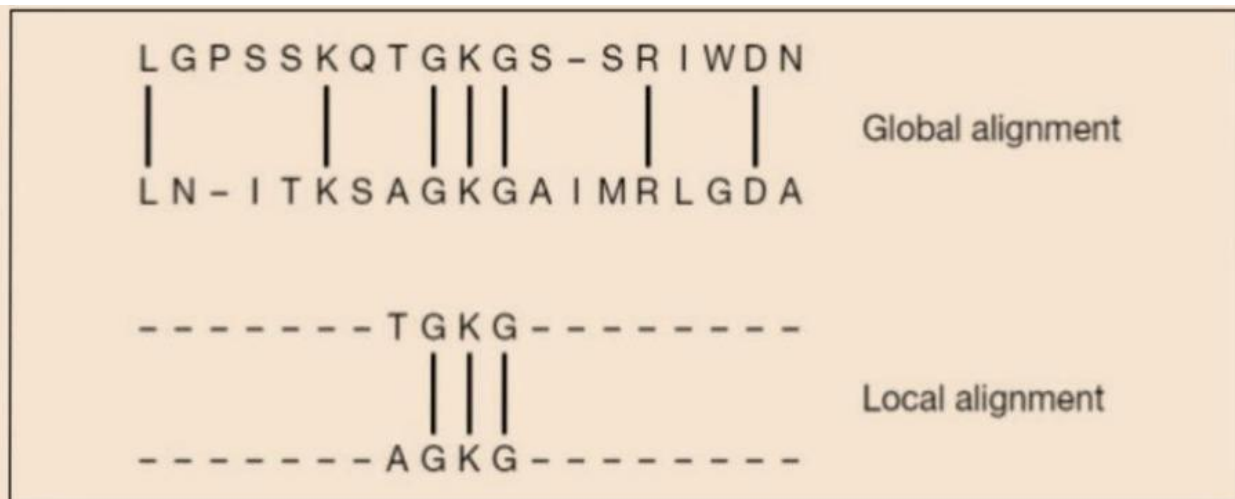
Pairwise: 2 sequences MSA: more than 2 sequences

F	G	K	-	G	K	G
F	G	K	F	G	K	G

F	G	K	-	G	K	G
F	G	K	F	G	K	G
-	G	K	Q	G	K	G
-	-	K	F	G	K	G

TYPES OF PAIRWISE ALIGNMENT

Feature	Global Alignment	Local Alignment
Scope	End-to-end alignment	Region-of-similarity alignment
Best Used For	Closely related sequences	Divergent or multi-domain sequences
Sequence Lengths	Must be similar in length	Can vary wildly in length
Primary Algorithm	Needleman-Wunsch	Smith-Waterman
Example Goal	Comparing human vs. chimp hemoglobin	Finding a conserved domain in a long genome



GLOBAL ALIGNMENT

- The alignment is stretched over the entire sequence length to include as many matching amino acids as possible upto and including the sequence ends. Vertical lines between the sequence indicate the presence of Identical amino acids.
- Involves the EMBOSS Needle tool.
- Needleman-Wunsch algorithm



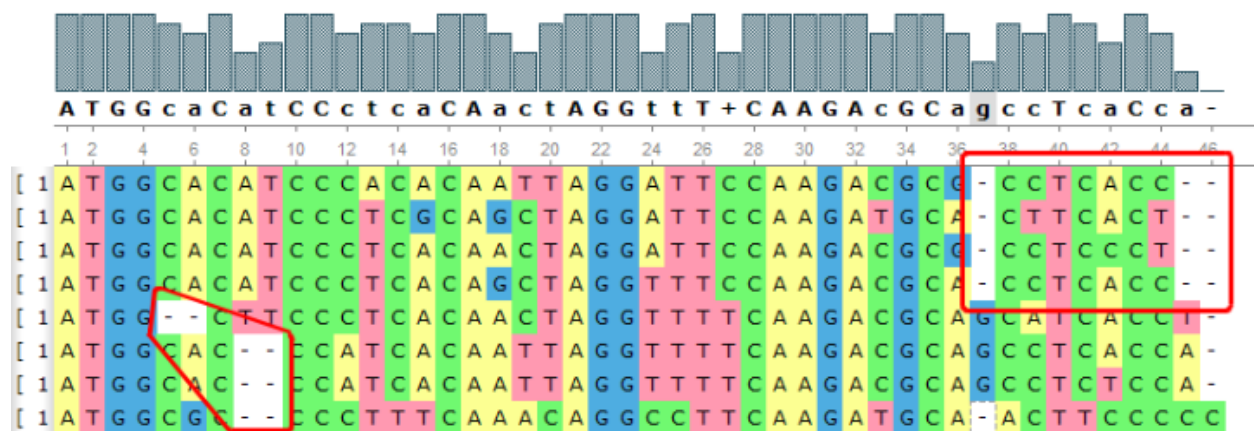
LOCAL ALIGNMENT

- The Alignment tends to stop at the end of the regions of identity or strong similarity. A much higher priority is given to finding these local regions than extending the alignment to include more neighboring amino acid pairs .
- Involves the BLAST tool.
- Smith-Waterman algorithm

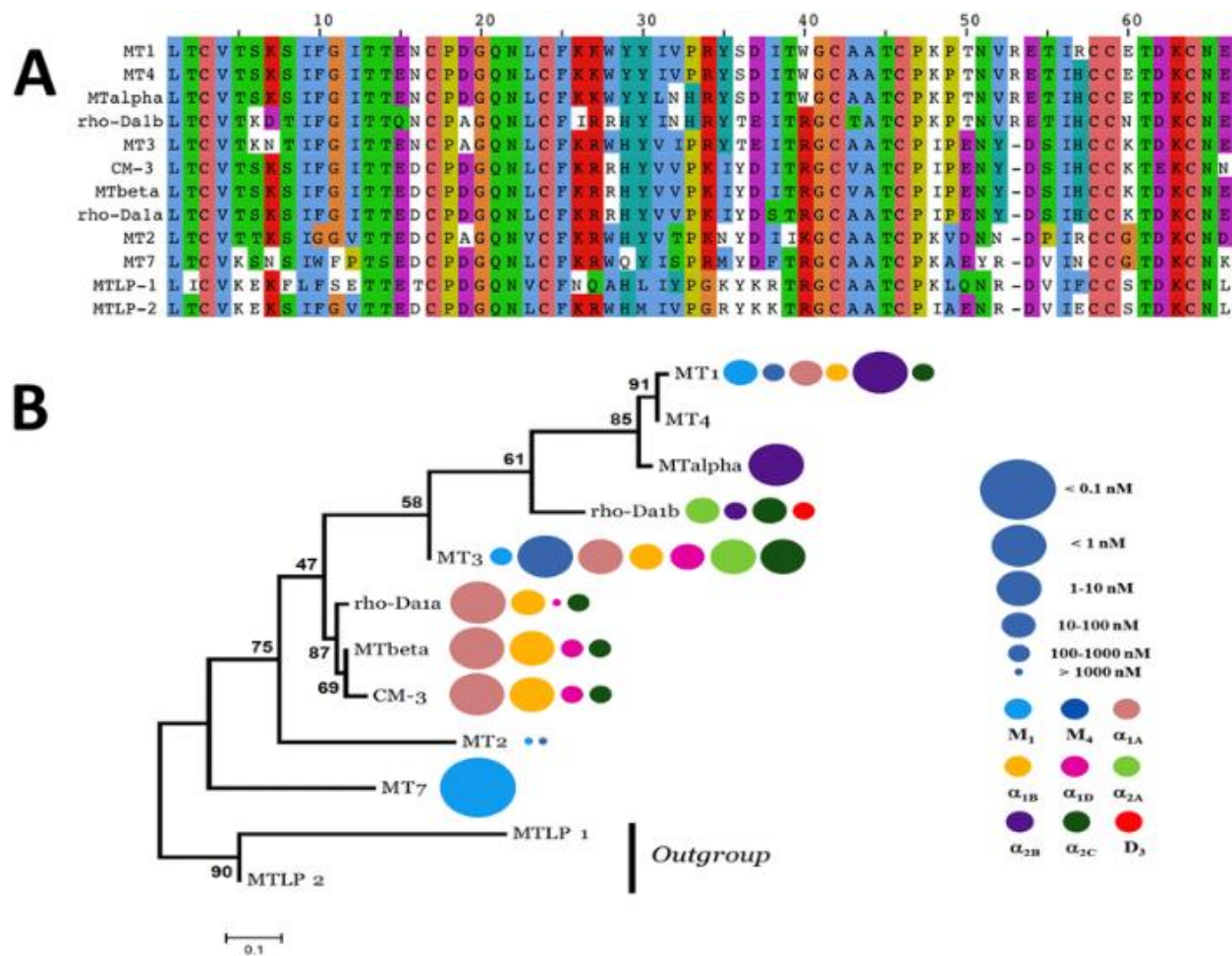


MULTIPLE SEQUENCE ALIGNMENT

- A multiple sequence alignment (MSA) is a sequence alignment of three or more biological sequences, generally protein, DNA, or RNA.
- In many cases, the input set of query sequences are assumed to have an evolutionary relationship by which they share a linkage and are descended from a common ancestor.



SEQUENCE ALIGNMENT IS USED FOR CREATING PHYLOGENETIC TREE



TOOLS INVOLVED IN ALIGNMENT

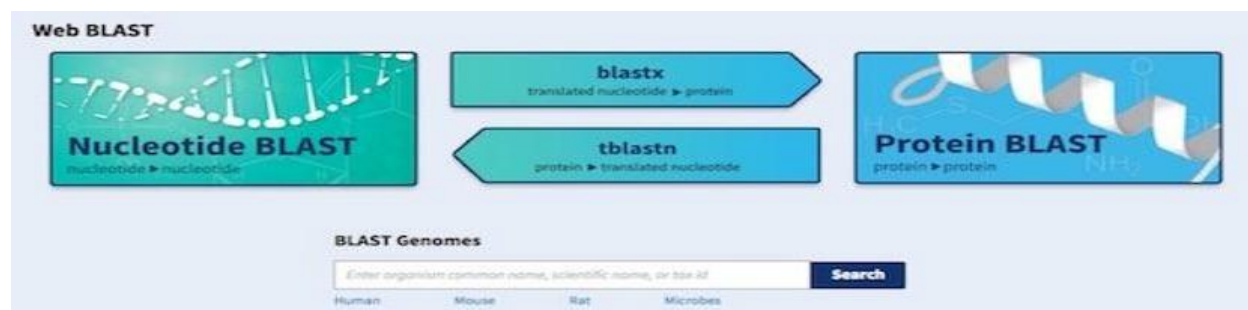
- MUSCLE
- M- view
- T-Coffee
- Clastal-W
- PyMOL
- SABERTOOTH
- Satsum

BLAST (Basic Local Alignment Search Tool)

- BLAST compares your sequence (the query) against sequences stored in a database and finds the closest matches.
- BLAST searches a massive database and returns similar sequences.
- BLAST is a set of sequence comparison algorithms used to search databases for optimal local alignments to a query.
- It breaks the query and databases sequences into fragments and seeks matches between them.
- Generally the alignments were done by full alignments by using dynamic programming.
- BLAST is 50 times faster than dynamic programming.

TYPES OF BLAST

BLAST Tool	Input	Searches Against	Use
BLASTn	DNA	DNA	Compare nucleotide sequences
BLASTp	Protein	Protein	Compare protein sequences
BLASTx	DNA	Protein	Translate DNA and search proteins
tBLASTn	Protein	Translated DNA	Protein vs genomes
tBLASTx	DNA	Translated DNA	Compare cross-species translated DNA



IMPORTANT BLAST TERMS

Core alignment metrics

- **Query:** The input sequence you provide to search against the database.
- **Subject:** The matching sequence found inside the database (also called the target).
- **Identity:** The percentage of exact character matches between the query and the subject.
- **Positivity:** The percentage of amino acids that are either identical or physically similar in structure.
- **Gaps:** Spaces introduced into an alignment to compensate for insertions or deletions during evolution.

Statistical significance terms

- **Raw score:** A numeric value calculated by adding match rewards and subtracting mismatch/gap penalties.
- **Bit score:** A normalized version of the raw score that accounts for database size, allowing you to compare results across different searches. Higher values indicate a stronger alignment.
- **E-value:** The **Expectation value**. It estimates the number of times a match this good would happen purely by random chance. Lower values (close to zero, like 10^{-50}) mean the match is statistically significant.
- **Low complexity region:** Sequences with repetitive tracts (like AAAAAA) that are filtered out because they cause false-positive matches.

Mathematical relationship of blast scores

The E-value (E) is mathematically derived from the bit score (S') and the search space size (N, which is the query length multiplied by the total database length) using the formula:

$$E = N \cdot 2^{-S'}$$

Because the bit score appears as a negative exponent, a linear increase in your **bit score** results in an exponential decrease in your **E-value**.

FIRST BLAST SEARCH

1. Go to the NCBI BLAST page.
2. Choose Nucleotide BLAST (BLASTn).
3. Open your insulin_dna.fasta.
4. Copy only the sequence (or paste the FASTA entry).
5. Paste it into the query box.
6. Leave most settings as default.

7. Click BLAST.
8. Wait a minute or two for the results.

RUN BLASTp

Now repeat the process with your insulin protein sequence using Protein BLAST (BLASTp).

Compare the top matches with the nucleotide search.